

2021 F=ma Exam: Problem 14

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From dimensional analysis, recall

$$[F] = \frac{ML}{T^2}$$

$$[R] = L$$

$$[v] = \frac{L}{T}$$

$$[\eta] = \frac{M}{LT}$$

We have

$$F \propto R^\alpha v^\beta \eta^\gamma$$
$$\frac{ML}{T^2} = L^\alpha \left(\frac{L}{T}\right)^\beta \left(\frac{M}{LT}\right)^\gamma$$

Counting powers of each dimension,

$$1 = \gamma$$

$$1 = \alpha + \beta - \gamma$$

$$-2 = -\beta - \gamma$$

so $\gamma = 1, \beta = 1, \alpha = 1$ and

$$F \propto Rv\eta$$

For the same F, η , we have

$$v \propto \frac{1}{R}$$

$$v' = \frac{R}{R'}v = \frac{1 \mu\text{m}}{0.5 \mu\text{m}}(20 \mu\text{m}) = 40 \mu\text{m/s}$$

so the answer is C.